INFORMATION RETRIEVAL (CS60092)

Programming Assignment III

Learning to Rank

Learning to Rank on the Cranfield Collection:
Boolean Weighted Zone Scoring and Linear-SVM Ranking

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Code repository

https://github.com/N-T-Raghava/Information-Retrival-CS60092

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Executive Summary

This report presents a reproducible learning-to-rank study on the Cranfield aerodynamics collection. The two learners required by the assignment were implemented: **Method A**, a Boolean weighted zone scorer whose four zone weights (title, author, bibliography, text) are learned by grid search, and **Method B**, a linear support vector machine that learns real-valued weights for 110 retrieval features computed on five document streams. A pairwise RankSVM (**Method B+**) was added as an extension, and untrained BM25 and TF-IDF cosine rankers serve as reference points. All learning used 180 training queries; the remaining 45 queries were held out, and a 5-fold cross-validation over all 225 queries quantifies how much the hold-out estimate can be trusted.

Principal result. On the held-out queries Method B reaches MAP **0.3134** against 0.2521 for Method A, a relative gain of 24.3% (the RankSVM reaches 0.3245). The difference is statistically supported (Wilcoxon $p = 0.023$; 95% bootstrap interval of the MAP difference $[+0.013, +0.116]$), and Method B wins on 27 / 2 / 16 queries (win / tie / loss). Five-fold cross-validation confirms the ordering: 0.324 for Method B against 0.270 for Method A.

Reference result. The learned SVM is, however, *not* better than untrained BM25: 0.3134 versus 0.3102 on the hold-out set ($p = 0.328$) and 0.324 versus 0.321 in cross-validation, a gap an order of magnitude smaller than the fold-to-fold standard deviation. The learned model matches a well-tuned lexical scorer and is somewhat ahead early in the ranking (P@1 0.422 versus 0.311), but that advantage is not statistically established with 45 queries.

Diagnostic on Method A. Combining zones is worth a great deal (MAP 0.192 for the title and 0.176 for the text alone, 0.2521 combined), yet tuning the weights adds little over equal weights (0.246). The large learned weight on the author zone (0.27) is not evidence that authors matter: that zone matches almost no query, so its weight only breaks ties. This is reported as a finding about the method and not over-interpreted.

Table 1: Summary of principal outcomes (45 held-out queries unless stated)

Outcome	Recorded finding
Hold-out MAP, Method A (zones)	0.2521
Hold-out MAP, Method B (linear SVM)	0.3134 (+24.3% over Method A)
Hold-out MAP, Method B+ (RankSVM)	0.3245 (+28.7% over Method A)
Untrained BM25 reference	0.3102
B vs. A, paired test	Wilcoxon $p = 0.023$; MAP difference +0.061, 95% CI [+0.013, +0.116]
B vs. BM25, paired test	Wilcoxon $p = 0.328$; difference +0.003, 95% CI [−0.035, +0.040] (not significant)
5-fold CV MAP (225 queries)	A 0.270; B 0.324; B+ 0.325; BM25 0.321
Best F1 at tuned cut-off k^*	B 0.283 ($k^* = 8$); BM25 0.273; A 0.234
Recall ceiling of lexical candidates	95.4% of relevant pairs share ≥ 1 query term

The assignment requires a report covering group details, problem statement, implementation details, plan of experiments, results and a discussion of results, together with program files and result files for the test queries. This report follows that structure; the supporting programs and result files are described in Appendix B.

1 Problem Statement and Objectives

The task is to rank the 1,400 Cranfield documents for each query with machine-learned scoring functions and to evaluate the rankings against human relevance judgements with precision, recall and F1. The central research question is:

How much can learned weights over zone-level and term-level evidence improve ranked retrieval compared with fixed lexical scorers, and does a linear SVM over rich features outperform Boolean weighted zone scoring?

The study stays inside the assignment boundary. Only lexical evidence is used: no dense embeddings, neural retrievers, external feedback or external relevance data. BM25 and TF-IDF cosine appear twice – as untrained reference rankers and as input features of the SVM – so that the contribution of *learning* can be separated from the contribution of the features themselves.

1.1 Objectives

- Implement Boolean weighted zone scoring and learn its four zone weights by grid search against the relevance judgements (Method A).
- Implement the 22 retrieval features on five document streams and learn their weights with a linear SVM (Method B); add a pairwise RankSVM as an extension (Method B+).
- Evaluate every system with the same qrels and metrics on queries not used for learning: precision, recall, F1, MAP and nDCG.
- Determine which streams and features carry the signal (ablations) and how stable the learned weights are.
- Test whether differences between systems are statistically meaningful given only 45 held-out queries.
- Keep every step deterministic and reproducible, and provide a program that ranks unseen queries with the final models.

2 Collection and Data Preparation

2.1 Collection Profile

Table 2: Cranfield collection and the experimental setup

Resource	Count / description
Documents	1,400
Queries	225 (every query has at least one relevant document)
Raw qrel records	1,837 (including 225 records with code −1)
Positive qrel records	1,612 (codes 1–4), i.e. 7.16 relevant documents per query
Document zones	title (`.T`), author (`.A`), bibliography (`.B`), text (`.W`)
Vocabulary	6,490 stems after stop-word removal and Porter stemming
Training / test queries	180 / 45 (random split, seed 42)
Candidate pairs	63.9% of all query–document pairs

Documents contain an identifier, a title, an author field, a bibliography field and an abstract (the text). Queries carry an identifier and the query text. The relevance judgements use codes 1–4 for increasingly marginal relevance and −1 for "no interest"; one −1 record exists for each query.

2.2 Parser Behavior and Data Quirks

Collection-specific details that can silently corrupt an experiment were handled explicitly:

1. **Repeated field markers.** Repeated `.A` or `.W` lines are accumulated, not overwritten.
2. **Positional query numbers.** `cranqrel` numbers the queries $1, \ldots, 225$ in file order, whereas the `.I` identifiers of `cran.qry` are not consecutive (001, 002, 004, ...). The mapping was verified empirically: BM25 over the whole document reaches MAP 0.321 with the positional numbering and 0.010 with the raw `.I` identifiers.
3. **Relevance definition.** Codes 1–4 are relevant, code −1 and unjudged pairs are not relevant, giving 1,612 relevant pairs.
4. **Encoding.** Files are read as Latin-1 so that no byte is lost, and CRLF line endings are stripped.

2.3 Preprocessing Chain

All text, for documents and queries alike, passes through the same analyser: lower-casing, extraction of alphanumeric tokens, English stop-word removal, removal of single letters, and Porter stemming. This yields a vocabulary of 6,490 stems. Four *zone* streams are built (title, author, bibliography, text) plus a fifth stream, *all*, that concatenates them. Because the text zone of Cranfield begins with the title, the title is effectively counted twice in the whole-document stream.

3 Implementation Details

The code is split into a shared library (`ir_lib.py`: parsing, analysis, features, metrics), the two learners (`method_a.py`, `method_b.py`), an experiment driver (`run_experiments.py`) and two utilities (`predict.py`, `evaluate.py`).

3.1 Method A: Boolean Weighted Zone Scoring

For a query q and document d, the Boolean match score of zone $i \in \{\text{title}, \text{author}, \text{bib}, \text{text}\}$ is the fraction of query terms that occur in that zone:

$$s_i(q,d) = \frac{|\{t \in q : \ t \in \text{zone}_i(d)\}|}{|q|}, \qquad \text{score}(q,d) = \sum_{i=1}^{4} g_i \, s_i(q,d), \quad g_i \geq 0, \ \sum_i g_i = 1. \tag{1}$$

For one-term queries s_i is exactly the 0/1 Boolean match; for longer queries it is the fraction of matched terms. The weights are learned by **exhaustive grid search** on the training queries. A coarse grid with step 0.05 covers the weight simplex with $\binom{23}{3} = 1{,}771$ points. A fine grid with step 0.01 in a ± 0.05 neighbourhood of the best coarse point ($11^3 = 1{,}331$ points) then refines the result, so that 3,102 weight vectors are evaluated for the MAP objective. Three objectives were compared: mean average precision (fixed in advance as the main choice), the textbook squared error $\frac{1}{n}\sum(\text{score} - \text{rel})^2$, and F1@10. Documents with equal scores are ranked in collection order, a deterministic tie-break.

3.2 Method B: Features and Linear SVM

For every query–document pair 22 features are computed on each of the five streams, giving 110 features. With $N = 1{,}400$ documents, tf the term count in the stream, L the stream length and df the stream document frequency:

Table 3: The 22 features computed on each of the five streams (110 features per query–document pair)

Feature (per stream)	**Definition used**
query term number / ratio	covered query terms / that number divided by $\lvert q\rvert$
stream length	L, number of tokens in the stream
idf	sum over covered query terms of $\ln \frac{N+1}{df+0.5}$
sum, min, max, mean, variance of tf	over the query terms (absent terms count as 0)
same for length-normalised tf	tf/L
same for tf·idf	$tf \times idf(t)$
Boolean model	1 if every query term occurs in the stream (AND)
vector space model	cosine of $(1+\ln tf)\cdot idf$ vectors of query and stream
BM25	Okapi BM25, $k_1 = 1.2$, $b = 0.75$, stream statistics

The BM25 feature is

$$\mathrm{BM25}(q,d) = \sum_{t\in q} \ln\!\Big(1+\frac{N-df+0.5}{df+0.5}\Big)\,\frac{tf\,(k_1+1)}{tf+k_1\big(1-b+b\,L/\bar{L}\big)}, \qquad k_1 = 1.2,\ b = 0.75. \quad (2)$$

The learner is the linear SVM of scikit-learn (`LinearSVC`, hinge loss, L2 regularisation, dual coordinate descent, at most 300 iterations) on standardised features, with balanced class weights. The learned weight vector w and intercept b give the ranking score $w\cdot x(q,d)+b$ for every document. Training uses all relevant pairs and a random sample of 150 non-relevant candidates per query, which keeps training fast while `class_weight=balanced` corrects the remaining imbalance. The regularisation constant $C \in \{0.001, 0.01, 0.1, 1\}$ is selected by 5-fold cross-validation *grouped by query* on the training queries (criterion: MAP).

3.3 Method B+: RankSVM

The RankSVM variant learns from pairs instead of single documents. For each training query, every relevant candidate d^+ is paired with 150 sampled non-relevant candidates d^-, and a linear SVM without intercept is trained to separate the difference vectors $x(q,d^+) - x(q,d^-)$ from their negatives (random sign flips keep the two classes balanced). It optimises the ordering of documents within a query directly, which is closer to the evaluation measure than pointwise classification.

3.4 Candidate Documents

Only *candidate* documents – those sharing at least one query term with the query in the whole-document stream – are scored by the SVMs; all others are ranked last. Candidates make up 63.9% of all pairs and contain 95.4% of the relevant pairs. This is therefore an upper bound on recall for every system here that depends on a shared query term.

3.5 Retrieved Set and Cut-off

Precision, recall and F1 require a retrieved *set*. They are reported at fixed cut-offs $k \in \{1, 5, 10, 20\}$ and at one tuned cut-off k^* per system, defined as the $k \in [1, 100]$ that maximises mean F1 on the training queries (on out-of-fold scores for the SVMs, so that k^* is not tuned on data the model was fitted on). All measures are macro-averages over queries.

3.6 Evaluation Implementation

All systems produce a score for every query–document pair and are evaluated by one shared function. Per-query AP is $\frac{1}{|\text{Rel}|} \sum_r P@r \cdot \mathbb{1}[d_r \text{ relevant}]$ and nDCG@10 uses binary gain. Statistical comparisons use the paired two-sided Wilcoxon signed-rank test on per-query AP, a percentile bootstrap (10,000 resamples, seed 0) of the mean paired AP difference, and win/tie/loss counts.

4 Plan of Experiments

The 225 queries are split at random (seed 42) into 180 training and 45 test queries. All learning — grid search, C, the standardiser and k^* — uses the training queries only; every number labelled "hold-out" is measured on the 45 test queries, which are listed in Appendix A.4.

Table 4: Experimental stages

Stage	Purpose	Search space
E1 Reference systems	Untrained lower and upper reference points	BM25, TF-IDF cosine, equal zone weights, each single zone
E2 Method A	Learn zone weights by grid search	3 objectives × (1,771 coarse + 1,331 fine) grid points
E3 Method B, C	Regularisation of the linear SVM	$C \in \{0.001, 0.01, 0.1, 1\}$, 5-fold query-grouped CV
E4 Ablation	Which streams carry the signal	text-zone only / no author–bib / all 110 features
E5 Method B+	Pairwise (RankSVM) training	same features, same C grid
E6 Significance	Are differences real?	Wilcoxon, paired bootstrap, win/tie/loss
E7 Cross-validation	Variance of the hold-out estimate	5 folds over all 225 queries
E8 Final models	Models for unseen queries	retrain on all 225 queries

Table 5: Evaluation metrics and interpretation

Metric	What it measures
MAP	Mean average precision over queries; primary effectiveness summary
P@k / R@k / F1@k	Precision, recall and their harmonic mean in the first k documents
k^* and F1@k^*	Cut-off that maximises mean F1 on the training queries, and the test F1 at that cut-off
nDCG@10	Rank-discounted gain of the first 10 documents (binary gain)
Wilcoxon, bootstrap	Paired tests on per-query AP

4.1 Selection Principle

The main objective of Method A (MAP) and the main variant of Method B (the pointwise SVM, as specified in the assignment) were fixed before any test result was inspected. The other objectives, the ablations and the RankSVM are reported for analysis and are not used to select the headline systems. Hyper-parameters (C, k^*) are chosen by cross-validation or out-of-fold scores on the training queries; C is selected within a small grid, so selection is deliberately coarse.

4.2 Reproducibility Note

All reported values are those recorded in `results/results.json` by the final run of `run_experiments.py`. The SVM solver is run with an iteration cap, so weights can differ slightly between platforms: an independent run in a different environment gave a Method B hold-out MAP of 0.316 against the recorded 0.3134. Differences of this size should not be interpreted. Further notes are in Appendix B.

5 Results – Reference Systems and Method A

5.1 Untrained References and Single Zones

Table 6: Hold-out results for the untrained references and single zones (45 queries)

System	MAP	nDCG@10	P@10	R@10	F1@10	P@k*	R@k*	F1@k*	k^*
TF-IDF cosine (whole doc)	0.298	0.370	0.220	0.372	0.252	0.253	0.353	0.267	8
BM25 (text zone)	0.306	0.382	0.227	0.399	0.263	0.253	0.361	0.269	8
BM25 (whole document)	0.310	0.385	0.224	0.391	0.261	0.293	0.319	0.273	6
Zones, equal weights (0.25)	0.246	0.328	0.176	0.335	0.211	0.203	0.319	0.225	8
Single zone: title	0.192	0.251	0.133	0.263	0.165	0.171	0.220	0.182	7
Single zone: text	0.176	0.235	0.138	0.234	0.158	0.175	0.217	0.174	7

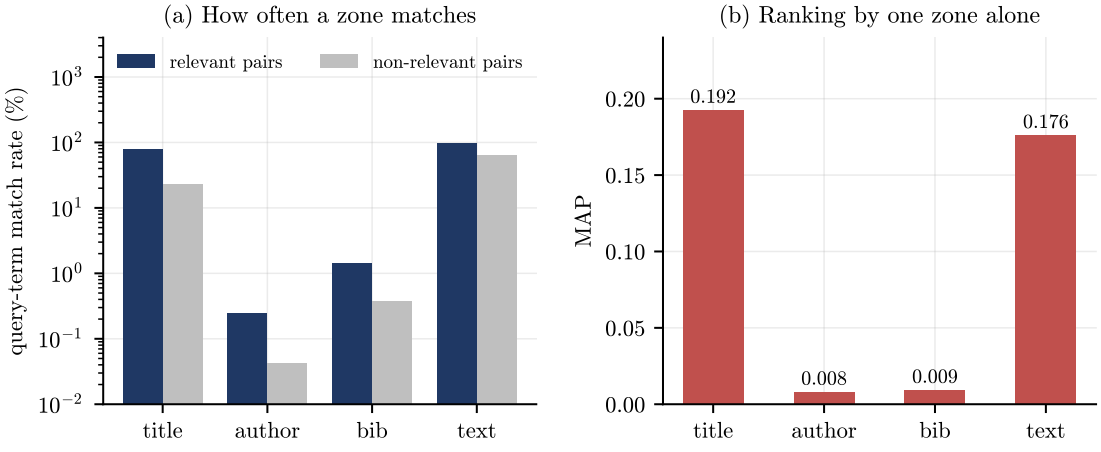


Figure 1: Evidence carried by each zone: (a) fraction of query–document pairs in which the zone matches at least one query term (logarithmic axis), (b) MAP when documents are ranked by one zone only.

Interpretation. BM25 and TF-IDF cosine on the whole document form a strong untrained baseline (MAP about 0.30–0.31). Zones differ greatly in how much evidence they carry. The title and text zones match 79% and 95% of relevant pairs against 23% and 64% of non-relevant pairs,

which is why they are informative. The author and bibliography zones match fewer than 2% of relevant pairs and ranking by them alone is essentially random (MAP below 0.01). They therefore cannot be expected to contribute to a combined score except as tie-breakers.

5.2 Zone Weights Learned by Grid Search

Table 7: Zone weights learned by grid search on the 180 training queries

Objective	title	author	bib	text	train value	test MAP
MAP	0.35	0.27	0.13	0.25	0.2750	0.252
Squared error (value = −SSE)	0.14	0.78	0.06	0.02	-0.0051	0.216
F1@10	0.30	0.35	0.15	0.20	0.2517	0.249

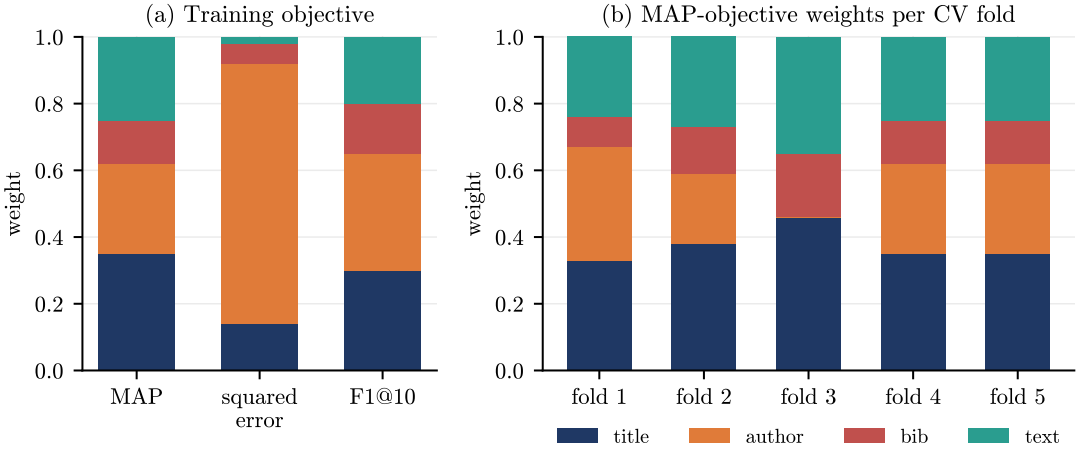


Figure 2: Zone weights learned by Method A: (a) for the three training objectives, (b) for the MAP objective in each cross-validation fold.

Table 8: Query-term match rate of every zone and mean zone score (all 225 queries)

Zone	match rate, relevant	match rate, non-relevant	mean score, relevant	mean score, non-relevant
title	78.8%	23.3%	0.2125	0.0324
author	0.2%	0.0%	0.0002	0.0000
bib	1.4%	0.4%	0.0022	0.0005
text	95.4%	63.7%	0.4042	0.1282

The MAP objective selects $(0.35, 0.27, 0.13, 0.25)$ for (title, author, bibliography, text). The squared-error objective selects a degenerate solution that puts 0.78 on the author zone. With only about 0.5% of the query–document pairs relevant, squared error is dominated by the non-relevant pairs, and it is minimised by the zone whose scores are closest to zero. The resulting ranking (MAP 0.216) is clearly worse, so the textbook criterion is a poor choice for this collection. The F1@10 objective behaves like MAP (0.249).

Stability of the weights. Panel (b) of Figure 2 shows that the weights found by the same grid search vary strongly across folds, in particular for the author zone, which ranges from 0.00 to 0.34. The five best grid points of the final training run differ by less than 10^{-5} in training MAP while

their author weight spans 0.20–0.27. Only the balance between title and text is actually identified by the data.

6 Results – Method B

6.1 Choice of the Regularisation Constant

Table 9: Mean CV MAP on the training queries for every value of C (the selected value is in bold)

Learner	$C = 0.001$	$C = 0.01$	$C = 0.1$	$C = 1$
Pointwise SVM (Method B)	**0.3112**	0.3097	0.2988	0.2766
RankSVM (Method B+)	**0.3159**	0.3127	0.2990	0.2915

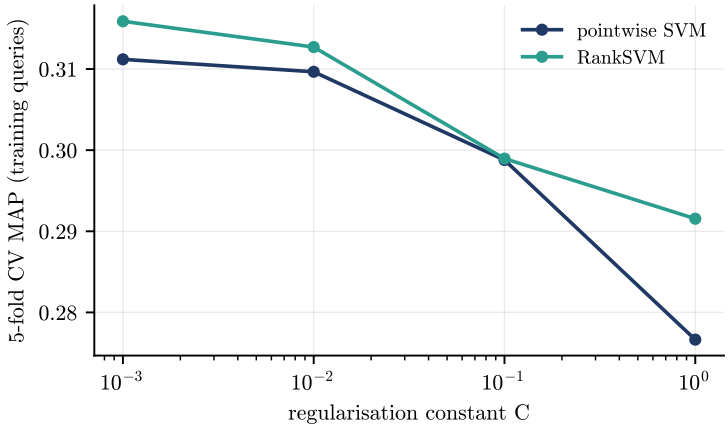


Figure 3: Cross-validation MAP on the training queries as a function of the SVM regularisation constant C.

Cross-validation MAP falls steadily as C grows for both learners, so the smallest value in the grid ($C = 0.001$) was selected for each. Strong regularisation is plausible here: many of the 110 features are highly correlated (minimum, maximum, mean and variance of related quantities). Because the optimum sits at the boundary of the grid, a still smaller C may be slightly better; this was not explored.

6.2 Feature-Set Ablation

Table 10: Method B feature-set ablation and the RankSVM variant (45 queries)

System	MAP	nDCG@10	P@10	R@10	F1@10	P@k*	R@k*	F1@k*	k^*
Method B (text-zone features)	0.305	0.382	0.229	0.401	0.265	0.279	0.358	0.283	7
Method B (no author/bib features)	0.317	0.403	0.236	0.414	0.275	0.267	0.380	0.284	8
Method B (SVM, all features)	**0.313**	**0.403**	**0.236**	**0.422**	**0.275**	**0.267**	**0.378**	**0.283**	**8**
Method B+ (RankSVM)	0.325	0.406	0.244	0.418	0.282	0.267	0.376	0.282	8

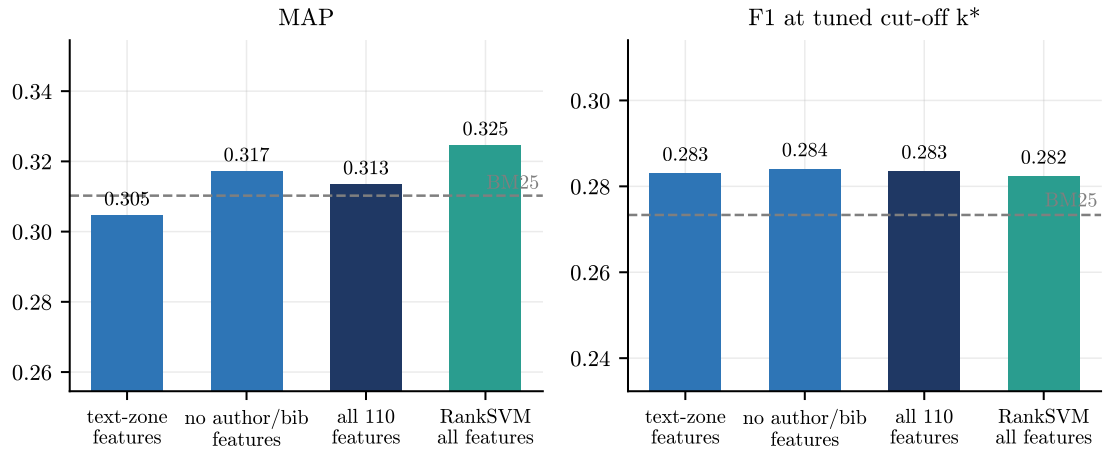


Figure 4: Effect of the feature set on Method B (dashed line: BM25) and the RankSVM variant.

Restricting the SVM to text-zone features gives MAP 0.305; adding the title and whole-document streams raises it to 0.3134 (and nDCG@10 from 0.382 to 0.403), whereas F1@k^* is unchanged at 0.283. Dropping the author and bibliography streams entirely gives 0.317, i.e. these streams contribute nothing, consistent with the behaviour of Method A. The small differences between these variants are within the noise of a 45-query test set.

6.3 What the SVM Learned

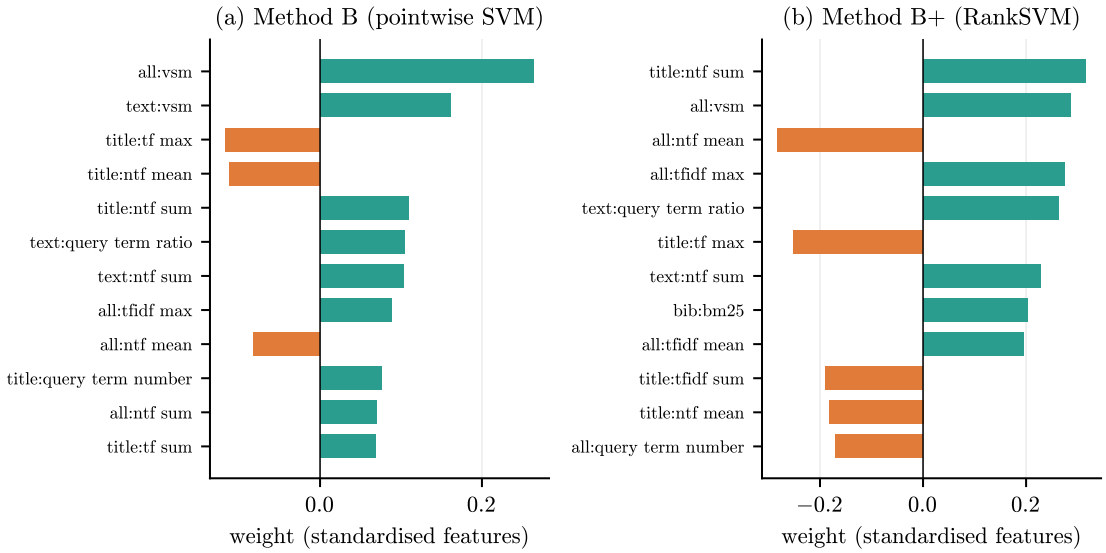


Figure 5: The twelve largest weights of (a) Method B and (b) the RankSVM. Features are standardised, so magnitudes are comparable within a panel.

The largest weights of Method B belong to the cosine scores of the whole document and of the text zone, followed by normalised term-frequency features of the title and text zones. Several strongly correlated features receive weights of opposite sign (for instance the sum and the mean of the normalised term frequency in the title). Individual weights are therefore not interpretable on their own; only the combined score is. The RankSVM has larger weights spread more evenly over the

features (Appendix A.2), as expected from an objective that works on differences of feature vectors.

7 Results – Final Comparison

7.1 Hold-out Comparison

Table 11: Final hold-out comparison on the 45 test queries (Method B, the required SVM method, in bold)

System	MAP	nDCG@10	P@10	R@10	F1@10	P@k*	R@k*	F1@k*	k^*
BM25 (whole document)	0.310	0.385	0.224	0.391	0.261	0.293	0.319	0.273	6
TF-IDF cosine (whole doc)	0.298	0.370	0.220	0.372	0.252	0.253	0.353	0.267	8
Method A (MAP objective)	0.252	0.330	0.176	0.338	0.211	0.222	0.310	0.234	7
Method B (SVM, all features)	**0.313**	**0.403**	**0.236**	**0.422**	**0.275**	**0.267**	**0.378**	**0.283**	**8**
Method B+ (RankSVM)	0.325	0.406	0.244	0.418	0.282	0.267	0.376	0.282	8

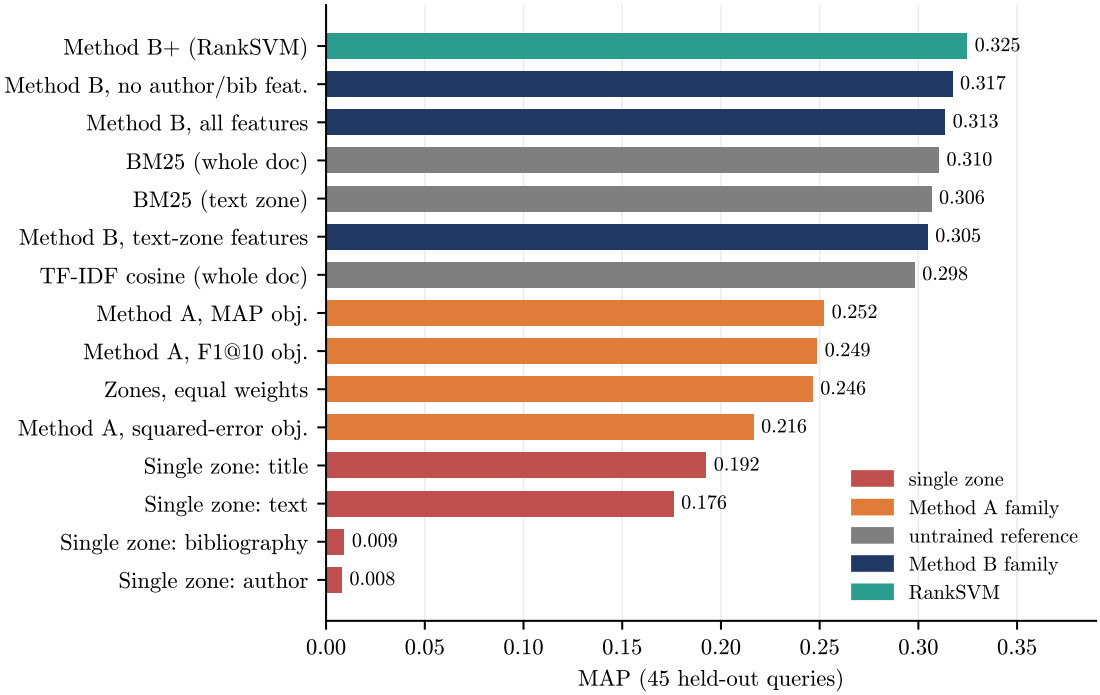


Figure 6: MAP of every system on the 45 held-out queries. Colours group the systems by family.

Method B improves on Method A on every reported measure, for instance P@10 from 0.176 to 0.236, R@10 from 0.338 to 0.422 and F1@k^* from 0.234 to 0.283. Against untrained BM25 the picture is different: Method B is ahead by a few percent in most measures (Figure 10) but the differences are small, and the paired tests of Section 8.2 do not separate them from zero. The RankSVM is nominally the best system (MAP 0.3245) but is statistically indistinguishable from Method B.

7.2 Precision, Recall and F1 at Fixed Cut-offs

Table 12: Precision, recall and F1 at fixed cut-offs (macro-average over 45 queries)

System	P@5	R@5	F1@5	P@10	R@10	F1@10	P@20	R@20	F1@20
Method A (MAP objective)	0.280	0.290	0.257	0.176	0.338	0.211	0.123	0.436	0.181
Method B (SVM, all features)	**0.311**	**0.294**	**0.271**	**0.236**	**0.422**	**0.275**	**0.152**	**0.535**	**0.222**
Method B+ (RankSVM)	0.333	0.316	0.291	0.244	0.418	0.282	0.159	0.543	0.229
BM25 (whole document)	0.307	0.285	0.265	0.224	0.391	0.261	0.154	0.528	0.221
TF-IDF cosine (whole doc)	0.324	0.300	0.279	0.220	0.372	0.252	0.148	0.514	0.213

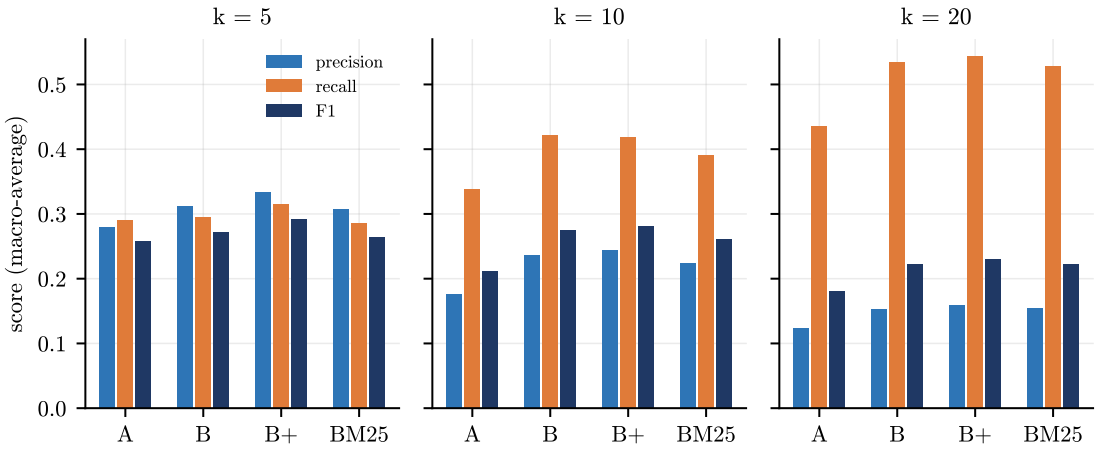


Figure 7: Precision, recall and F1 at three fixed cut-offs (A = Method A, B = Method B, B+ = RankSVM).

Precision decreases and recall increases with the cut-off, as expected. Around $k = 10$, Method B reaches a recall of 0.422 at a precision of 0.236; at $k = 20$ recall rises to 0.535 while precision falls to 0.152.

7.3 Precision–Recall and Cut-off Analysis

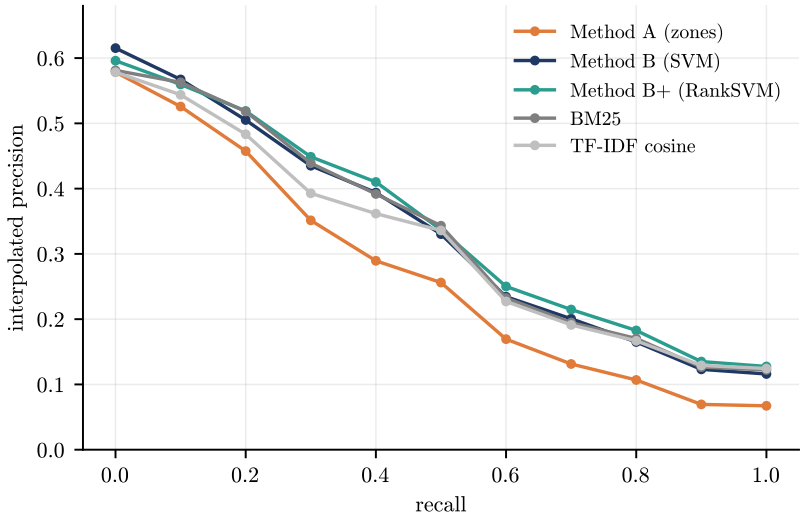


Figure 8: 11-point interpolated precision–recall curves on the 45 test queries.

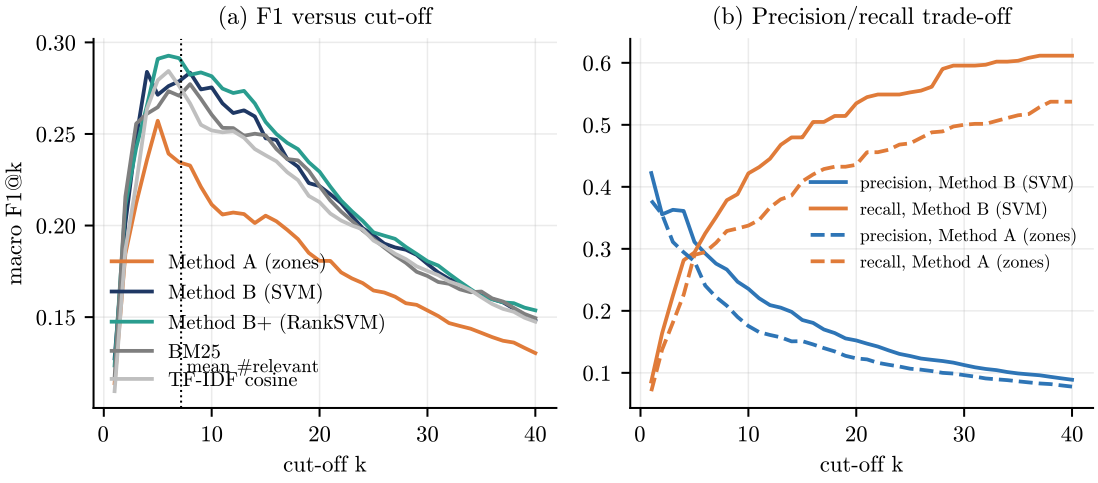


Figure 9: (a) Macro-F1 as a function of the cut-off k; the dotted line marks the mean number of relevant documents per query. (b) Precision and recall as functions of k.

Method A lies below all other systems over most of the recall range. The four stronger systems are close and their curves cross, which is another view of the fact that the learned combination is competitive with, but not clearly better than, BM25. F1 peaks at $k = 7$–8, close to the mean number of relevant documents per query (7.16); the tuned cut-offs are $k^* = 8$ for Method B, with precision 0.267, recall 0.378 and F1 0.283 (BM25: F1 0.273).

7.4 Relative Improvement

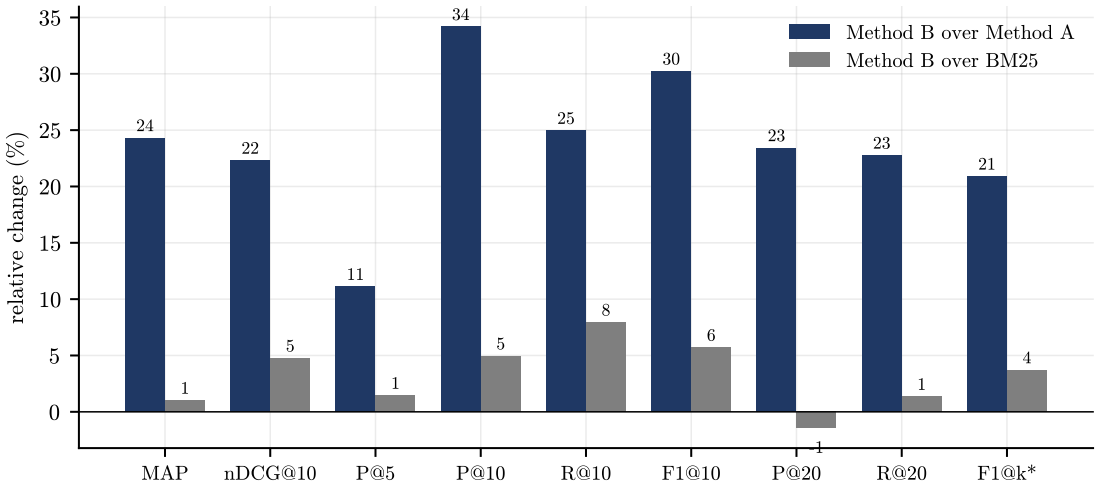


Figure 10: Relative change (%) of Method B over Method A and over BM25 across the reported metrics.

Relative to Method A the SVM gains between 11% (P@5) and 34% (P@10); relative to BM25 the gains are between −1% and +8%, the largest being recall at 10 (8%).

8 Generalisation, Significance and Efficiency

8.1 Five-Fold Cross-Validation

Table 13: Five-fold cross-validation over all 225 queries (mean ± standard deviation over the 5 folds of 45 queries)

System	MAP	nDCG@10	P@10	R@10	F1@10	F1@k^*
Method A	0.270 ± 0.032	0.352 ± 0.042	0.214 ± 0.023	0.366 ± 0.051	0.244 ± 0.024	0.246 ± 0.029
Method B (SVM)	**0.324 ± 0.025**	**0.404 ± 0.034**	**0.248 ± 0.026**	**0.414 ± 0.047**	**0.280 ± 0.025**	**0.283 ± 0.028**
Method B+ (RankSVM)	0.325 ± 0.029	0.400 ± 0.035	0.247 ± 0.027	0.409 ± 0.051	0.278 ± 0.025	0.283 ± 0.025
BM25 (whole document)	0.321 ± 0.023	0.395 ± 0.030	0.238 ± 0.026	0.396 ± 0.039	0.269 ± 0.022	0.278 ± 0.021

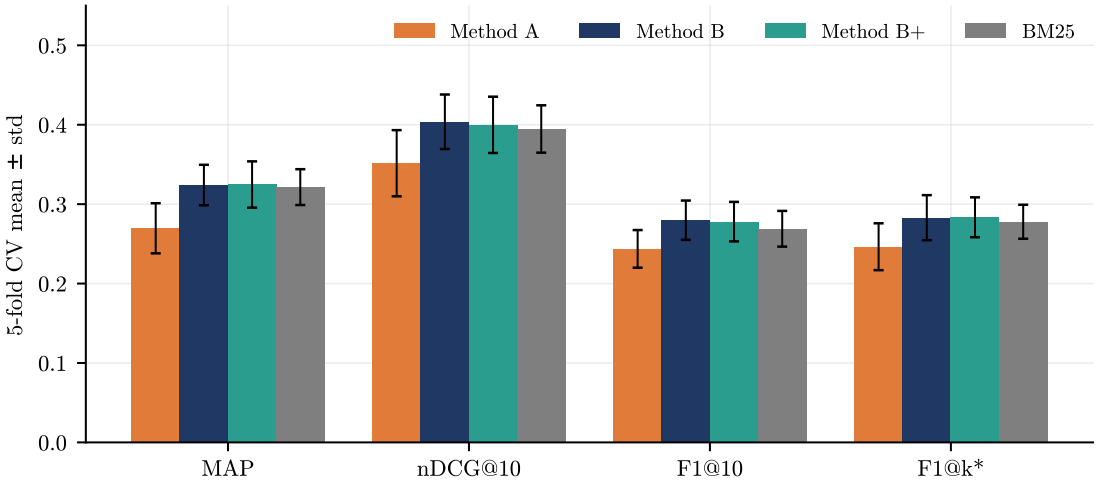


Figure 11: Five-fold cross-validation over all 225 queries (bars: mean, whiskers: standard deviation over folds).

Over all 225 queries, Method B (0.324) and the RankSVM (0.325) clearly outperform Method A (0.270), while BM25 (0.321) is level with them. The standard deviation over folds is about 0.025 for MAP: this is the size of difference that a single 45-query split cannot resolve, and the reason why the Method B–BM25 gap of 0.003 must not be interpreted.

8.2 Statistical Significance and Per-Query Behaviour

Table 14: Paired comparisons on per-query AP (45 test queries). W/T/L = queries where the first system wins / ties / loses

Comparison	ΔMAP	95% bootstrap CI	Wilcoxon p	W/T/L
Method B vs. Method A	+0.0613	[+0.013, +0.116]	0.023	27/2/16
Method B+ vs. Method A	+0.0724	[+0.019, +0.131]	0.019	28/2/15
Method B vs. BM25	+0.0031	[−0.035, +0.040]	0.328	27/2/16
Method B+ vs. BM25	+0.0143	[−0.021, +0.050]	0.391	25/2/18
Method B+ vs. Method B	+0.0111	[−0.007, +0.031]	0.512	24/3/18
Method A vs. text zone alone	+0.0762	[+0.027, +0.124]	0.0006	33/1/11
Method A vs. BM25	−0.0582	[−0.116, −0.008]	0.155	22/1/22

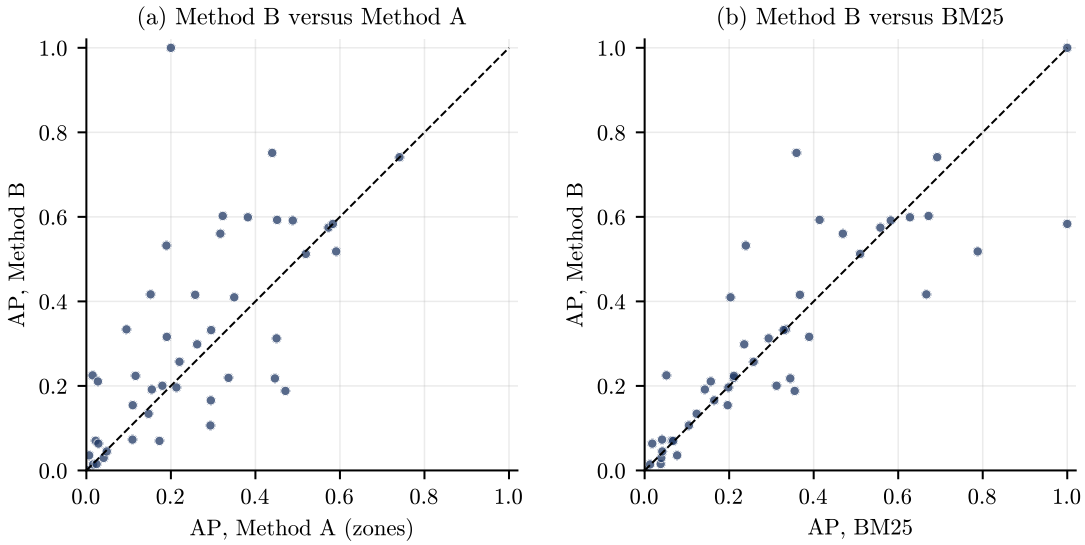


Figure 12: Per-query average precision of Method B against (a) Method A and (b) BM25. Points above the diagonal are queries on which Method B wins.

Method B beats Method A on 27 / 2 / 16 queries (win/tie/loss) and the improvement is significant ($p = 0.023$). Against BM25, Method B is ahead on 27 / 2 / 16 queries, yet the mean paired difference is only +0.003 with a bootstrap interval of [-0.035, +0.040] that comfortably includes zero ($p = 0.328$). The RankSVM and the pointwise SVM are not distinguishable ($p = 0.512$). Combining zones is significantly better than the best single zone ($p = 0.0006$ against the text zone).

Table 15: Early precision: P@1 (45 queries)

System	P@1	P@5	nDCG@10
Method A (MAP objective)	0.378	0.280	0.330
Method B (SVM, all features)	0.422	0.311	0.403
Method B+ (RankSVM)	0.400	0.333	0.406
BM25 (whole document)	0.311	0.307	0.385
TF-IDF cosine (whole doc)	0.356	0.324	0.370

8.3 Efficiency

Table 16: Recorded runtimes (wall-clock on the machine used for the final run)

Component	Recorded value
Index + features for 225 queries × 1,400 documents	≈ 2 s
Method A grid search, MAP objective	81.3 s (3,102 grid points)
Method A grid search, squared-error objective	31.9 s (2,482 grid points)
Method A grid search, F1@10 objective	78.6 s (3,102 grid points)
Method B: choose C (20 fits) + final fit + out-of-fold k^*	71.9 s
Method B ablations (single fit each)	1.6 s (text zone), 4.3 s (no author/bib)
Method B+ (RankSVM): choose C + final fit + out-of-fold k^*	275.9 s
Whole hold-out stage / 5-fold CV / final models	549 s / 488 s / 125 s (console log of the final run)

All learning is cheap. Feature extraction for all 315,000 query–document pairs takes about two seconds; one evaluation of a zone-weight vector takes about 26 ms, so the full grid search needs about 80 s. The pointwise SVM fits in a few seconds; the RankSVM is the most expensive step because of the larger pairwise training set and the 20 fits of the C search. Ranking a new query requires one dot product of 110 weights with the pair features of each candidate document.

9 Discussion of Results

9.1 Effect of Zone Weighting

Zones are complementary: the title and the text zone reach MAP 0.192 and 0.176 separately, but 0.2521 when combined, a difference that is significant against the text zone alone ($p = 0.0006$). Learning the weights, however, contributes little beyond this: equal weights already reach 0.246. The reason is structural. Only two of the four zones carry usable evidence, and for those two a 50:50 balance is already near the optimum; the surface is flat. The author and bibliography zones act purely as tie-breakers, and the weights assigned to them should not be read as importance. A further limitation is inherent in Boolean scoring: with few distinct score values many documents tie and are ordered arbitrarily, which plausibly explains part of the gap to BM25, although this was not isolated experimentally.

9.2 Effect of Learned Feature Weights

The SVM replaces binary matching by graded evidence – term frequencies, idf and length normalisation – and combines it across streams. This yields 24.3% higher MAP than Method A. Its performance matches BM25 because BM25 and the cosine score are themselves among the 110 features and receive the largest weights; with 1,612 positive pairs there is little left to learn beyond a well-tuned lexical score. The ablation shows that the title and whole-document streams add a small, noisy gain over the text zone.

9.3 Pointwise versus Pairwise Learning

The RankSVM is nominally better in MAP (0.3245 against 0.3134) and in P@10, but the difference is not significant ($p = 0.512$) and vanishes in cross-validation (0.325 against 0.324). Both models select the same strong regularisation. For this collection the choice between pointwise and pairwise training is therefore not critical.

9.4 Metric Trade-offs

Different measures favour different systems. TF-IDF cosine has the highest P@1 among the reference rankers (0.356) and Method B the highest overall (0.422), whereas BM25 and the SVMs are closer in MAP. Because queries have about seven relevant documents, F1 is maximised at a cut-off of 7–8; at larger cut-offs recall is bought with a rapid loss of precision (Figure 9b). Reporting several measures is therefore necessary when comparing systems.

9.5 Baseline Comparison

The untrained BM25 baseline is not beaten significantly by any learned system in this study. This does not make learning pointless: the SVM reproduces BM25-level quality from raw features with no hand-set parameters, and the learned weights can combine additional evidence. It does mean that on a small collection with 225 queries the features, not the learner, determine the ceiling. For context, the previous assignment's tuned SMART `lnc.ltc`+Rocchio system recorded an all-topic MAP of 0.3253; the cross-validated MAP of Method B here (0.324) is of similar size, but the figures are not directly comparable, because the two studies differ in the relevance threshold (grades 1–3 there, 1–4 here), tokeniser and topic split.

9.6 Limitations and Threats to Validity

- The hold-out set contains only 45 queries, and cross-validation shows a fold-to-fold MAP standard deviation of 0.02–0.03; differences below this are within noise.
- Relevance codes 1–4 are collapsed to binary relevance; graded relevance is not exploited.
- SVM training subsamples 150 negative candidates per query, and the solver stops after 300 iterations, so weights are approximate and slightly platform-dependent.
- C is at the boundary of its grid; k^* and C are selected greedily on the training queries.
- Boolean zone scores produce many ties, resolved by collection order.
- Result files keep the top 100 documents per query, so MAP recomputed from them with `evaluate.py` is slightly lower than in the tables (for Method B: 0.3075 instead of 0.3134).
- The final models in `models/` are trained on all 225 queries and cannot be evaluated on Cranfield itself; their expected quality is that of the cross-validation rows of Table 13.

10 Conclusion

This project delivers Boolean weighted zone scoring and a linear-SVM ranker for the Cranfield collection, together with a pairwise extension, shared evaluation logic, statistical tests and a program that ranks unseen queries. The central empirical findings are:

1. Learning real-valued weights over 110 features (Method B, MAP 0.3134) is significantly better than learning four Boolean zone weights (Method A, MAP 0.2521), a relative gain of 24.3%,

confirmed by 5-fold cross-validation (0.324 versus 0.270).

2. The SVM matches but does not significantly beat untrained BM25 (0.3102); the pairwise RankSVM is nominally best (0.3245) but not distinguishable from the pointwise SVM.
3. In Method A only the title–text balance is learned; the author and bibliography weights are not identifiable, and the squared-error objective is unsuitable.
4. All results rest on 45 held-out queries with a cross-validation standard deviation of about 0.025 MAP, which sets the resolution at which systems can be compared.

For unseen queries we recommend Method B or Method B+, whose cross-validated quality is about 0.32 MAP and F1 about 0.28 at the tuned cut-off.

A Detailed Recorded Results

The appendix preserves the complete numerical data behind the figures.

A.1 All Hold-out Results

Table 17: All hold-out results recorded by the final run (45 queries)

System	MAP	nDCG@10	P@10	R@10	F1@10	P@k*	R@k*	F1@k*	k^*
BM25 (text zone)	0.306	0.382	0.227	0.399	0.263	0.253	0.361	0.269	8
BM25 (whole document)	0.310	0.385	0.224	0.391	0.261	0.293	0.319	0.273	6
TF-IDF cosine (whole doc)	0.298	0.370	0.220	0.372	0.252	0.253	0.353	0.267	8
Zones, equal weights (0.25)	0.246	0.328	0.176	0.335	0.211	0.203	0.319	0.225	8
Single zone: title	0.192	0.251	0.133	0.263	0.165	0.171	0.220	0.182	7
Single zone: author	0.008	0.000	0.000	0.000	0.000	0.003	0.006	0.004	25
Single zone: bibliography	0.009	0.006	0.007	0.006	0.006	0.005	0.018	0.007	33
Single zone: text	0.176	0.235	0.138	0.234	0.158	0.175	0.217	0.174	7
Method A (MAP objective)	0.252	0.330	0.176	0.338	0.211	0.222	0.310	0.234	7
Method A (squared-error obj.)	0.216	0.285	0.164	0.309	0.198	0.181	0.285	0.204	8
Method A (F1@10 objective)	0.249	0.327	0.173	0.338	0.210	0.208	0.329	0.233	8
Method B (SVM, all features)	**0.313**	**0.403**	**0.236**	**0.422**	**0.275**	**0.267**	**0.378**	**0.283**	**8**
Method B (text-zone features)	0.305	0.382	0.229	0.401	0.265	0.279	0.358	0.283	7
Method B (no author/bib features)	0.317	0.403	0.236	0.414	0.275	0.267	0.380	0.284	8
Method B+ (RankSVM)	**0.325**	**0.406**	**0.244**	**0.418**	**0.282**	**0.267**	**0.376**	**0.282**	**8**

Table 18: All hold-out results at fixed cut-offs (45 queries)

System	P@5	R@5	F1@5	P@10	R@10	F1@10	P@20	R@20	F1@20
BM25 (text zone)	0.320	0.299	0.276	0.227	0.399	0.263	0.151	0.519	0.217
BM25 (whole document)	0.307	0.285	0.265	0.224	0.391	0.261	0.154	0.528	0.221
TF-IDF cosine (whole doc)	0.324	0.300	0.279	0.220	0.372	0.252	0.148	0.514	0.213
Zones, equal weights (0.25)	0.271	0.279	0.249	0.176	0.335	0.211	0.123	0.429	0.180
Single zone: title	0.209	0.201	0.194	0.133	0.263	0.165	0.101	0.374	0.151
Single zone: author	0.000	0.000	0.000	0.000	0.000	0.000	0.003	0.006	0.004
Single zone: bibliography	0.009	0.003	0.005	0.007	0.006	0.006	0.006	0.014	0.008
Single zone: text	0.200	0.181	0.171	0.138	0.234	0.158	0.093	0.342	0.136
Method A (MAP objective)	0.280	0.290	0.257	0.176	0.338	0.211	0.123	0.436	0.181
Method A (squared-error obj.)	0.240	0.230	0.218	0.164	0.309	0.198	0.112	0.395	0.165
Method A (F1@10 objective)	0.276	0.285	0.252	0.173	0.338	0.210	0.124	0.437	0.182
Method B (SVM, all features)	0.311	0.294	0.271	0.236	0.422	0.275	0.152	0.535	0.222
Method B (text-zone features)	0.333	0.315	0.290	0.229	0.401	0.265	0.151	0.533	0.219
Method B (no author/bib features)	0.324	0.312	0.285	0.236	0.414	0.275	0.153	0.533	0.223
Method B+ (RankSVM)	0.333	0.316	0.291	0.244	0.418	0.282	0.159	0.543	0.229

A.2 Largest SVM Weights

Table 19: The 15 largest weights of Method B (standardised features)

Rank	Feature (stream:name)	Weight
1	`all:vsm`	+0.264
2	`text:vsm`	+0.161
3	`title:tf_max`	−0.117
4	`title:ntf_mean`	−0.112
5	`title:ntf_sum`	+0.110
6	`text:query_term_ratio`	+0.104
7	`text:ntf_sum`	+0.103
8	`all:tfidf_max`	+0.089
9	`all:ntf_mean`	−0.082
10	`title:query_term_number`	+0.076
11	`all:ntf_sum`	+0.069
12	`title:tf_sum`	+0.068
13	`all:query_term_ratio`	+0.066
14	`all:tfidf_var`	+0.065
15	`title:ntf_max`	+0.054

Table 20: The 15 largest weights of Method B+ (RankSVM)

Rank	Feature (stream:name)	Weight
1	`title:ntf_sum`	+0.316
2	`all:vsm`	+0.286
3	`all:ntf_mean`	−0.283
4	`all:tfidf_max`	+0.275
5	`text:query_term_ratio`	+0.263
6	`title:tf_max`	−0.251
7	`text:ntf_sum`	+0.227
8	`bib:bm25`	+0.203
9	`all:tfidf_mean`	+0.196
10	`title:tfidf_sum`	−0.190
11	`title:ntf_mean`	−0.183
12	`all:query_term_number`	−0.170
13	`title:ntf_max`	+0.168
14	`title:idf`	−0.166
15	`title:tf_sum`	+0.150

A.3 Zone Weights per Cross-Validation Fold

Table 21: Zone weights found by the MAP-objective grid search in each CV fold

Fold	title	author	bib	text
1	0.33	0.34	0.09	0.24
2	0.38	0.21	0.14	0.27
3	0.46	0.00	0.19	0.35
4	0.35	0.27	0.13	0.25
5	0.35	0.27	0.13	0.25

A.4 Held-out Query Identifiers

The 45 test queries are, in the numbering of `cranqrel`: 10, 16, 17, 19, 26, 31, 46, 61, 69, 76, 80, 83, 87, 97, 102, 105, 112, 115, 118, 119, 121, 142, 143, 144, 145, 149, 153, 166, 168, 173, 176, 177, 179, 183, 185, 194, 196, 199, 201, 202, 208, 214, 215, 221, 223.

B Reproducibility and Implementation Walkthrough

B.1 Core Data Structures

The `Index` object holds one sparse term-frequency matrix per stream (title, author, bib, text, all), the stream lengths, idf values and document-vector norms. For a query it returns a 1400×110 feature matrix; stacking all 225 queries gives a $225 \times 1400 \times 110$ tensor (≈ 139 MB in single precision) from which both methods are trained. `SVMRanker` wraps the standardiser and the linear SVM (pointwise or RankSVM) and exposes `fit` and `score`.

B.2 Execution Flow

1. Load the three Cranfield files from a directory or `cran.tar.gz`; build the index; compute the feature tensor.
2. Split the queries into 180 training and 45 test queries (seed 42).
3. Evaluate the reference systems; run the grid search for the three objectives of Method A.
4. For each SVM variant choose C by 5-fold grouped cross-validation, fit on the training queries, obtain out-of-fold scores to tune k^*, and score the test queries.
5. Run Wilcoxon tests; write the TREC result files, the predicted-relevant files and `results.json`.
6. With `--cv`: repeat grid search, fitting and k^* selection inside each of 5 folds over all 225 queries.
7. Retrain Method A, Method B and Method B+ on all 225 queries and save them in `models/`.

B.3 Recommended Execution

```
pip3 install -r requirements.txt
python3 run_experiments.py --data data --cv            # about 20 minutes
python3 predict.py --data data --queries NEW.qry --method B --out results/new
python3 evaluate.py --run results/methodB_test_results.txt \
        --qrels data/cranqrel --k 5 10
cd report && python3 make_figures.py && python3 build_report.py
```

`predict.py` accepts `--method A|B|B-rank` and writes a TREC-format ranking (`qid Q0 docid rank score tag`, top 100) and the top-k^* documents as `qid docid` pairs. Queries are numbered $1, \ldots, n$ in file order, the convention of `cranqrel`; `--id-mode orig` keeps the `.I` identifiers instead.

B.4 Generated Artifacts

Table 22: Generated artifacts

Artifact	Purpose
`results/methodA_test_results.txt`	Method A ranking of the 45 test queries (TREC format, top 100)
`results/methodB_test_results.txt`	Method B (linear SVM) ranking of the test queries
`results/methodB_ranksvm_test_results.txt`	Method B+ (RankSVM) ranking of the test queries
`results/*_predicted_relevant.txt`	Top-k^* documents per query as `qid docid` pairs
`results/results.json`	All metrics, per-query AP, curves, weights, CV results
`models/method_A.json`	Final zone weights and k^* (trained on all 225 queries)
`models/method_B.pkl`	`models/method_B_ranksvm.pkl`: final SVM models, scaler and k^*
`report/figures/*.pdf`	All figures of this report